UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PROGRAMMING I
COURSE CODE	CSC402
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages.

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PART A (10 MARKS)

Answer ALL questions.

1. What is pseudocode?
   A. Machine code produced by an assembler
   B. A C++ header file
   C. A structured English-like description of an algorithm
   D. A type of compiler error

2. What is displayed by the following code?

```
for (int i = 1; i <= 4; i++)
    cout << i * 2;
```

   A. 2468
   B. 1357
   C. 246
   D. 1234

3. Which operator sends data to cout?
   A. ==
   B. <<
   C. =
   D. >>

4. Given the declaration, what is displayed?

```
char word[6] = {'p', 'r', 'i', 'n', 't', '\0'};
cout << word[3];
```

   A. i
   B. n
   C. t
   D. p

5. Which of the following declarations are valid?
   i. int a[5];
   ii. int b;
   iii. int c[3] = {1, 2, 3};
   iv. int d[];
   A. i and iii
   B. ii and iv
   C. iii and iv
   D. i and ii

6. Which statement immediately exits a loop?
   A. return 0
   B. cin
   C. continue
   D. break

7. Which statement is most suitable for selecting among several constant cases?
   A. while
   B. switch

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C. for
D. do...while

8. Which data type stores a whole number?
A. float
B. string
C. int
D. char

9. Which of the following are valid identifiers?
i. total_score
ii. 2total
iii. _count
iv. int
A. ii and iv
B. i and ii
C. i and iii
D. iii and iv

10. Which item is NOT a programming paradigm?
A. Debugging
B. Procedural programming
C. Structured programming
D. Object-oriented programming

PART B (60 MARKS)

QUESTION 1

This question tests high-level languages and debugging.

a) Define syntax error and logic error.

(2 marks)

b) Explain TWO advantages of high-level languages over machine language.

(3 marks)

QUESTION 2

A program reads a student's score from 0 to 100 and displays the grade shown in Table 1.

Table 1. Grade Scale

Score	Grade
80-100	A
60-79	B
50-59	C
40-49	D
0-39	F

a) Draw a flowchart for the grade classification.

(8 marks)

b) Write pseudocode for the algorithm and convert it into a complete C++ program.

(5 marks)

QUESTION 3

This question tests operator precedence and assignment.

a) Evaluate the following expressions using the declarations in the code block.

```
int a = 12, b = 5;
double c = 2.5;
```

(6 marks)

b) Write C++ statements to declare two integers a and b, read their values, and display their sum, difference, product and remainder.

(5 marks)

QUESTION 4

Study the following C++ program segment:

```
int total = 0;
for (int i = 1; i <= 5; i++) {
    total += i;
}
cout << total;
```

a) Trace the program and state the output.

(6 marks)

b) Rewrite the loop using a while loop. Then explain what changes if the condition is i < 5.

(9 marks)

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QUESTION 5

A shop stores the item prices shown in Table 1.

a) Declare an array named prices using the values in Table 1. Display all values.

Table 2. Item Prices

Item	Price (RM)
1	20.50
2	55.00
3	15.90
4	75.00
5	10.00

(6 marks)

b) Write a program segment that applies a 10% discount to prices above RM50 and stores the results in a second array. Display both arrays.

(10 marks)

PART C (30 MARKS)

QUESTION 1

A lecturer records the marks of ten students in an array.

Write a complete C++ program that reads ten marks, displays the average, the highest mark and its index, and counts how many marks are above the average.

(15 marks)

QUESTION 2

A bookstore records up to five books. Table 1 shows an example of the fields.

Table 3. Book Record Fields

Title	Price (RM)	Quantity
Book A	45.00	3
Book B	30.00	5
Book C	55.00	2

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

(15 marks)

END OF QUESTION PAPER

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